Response to the In-house Editor and Reviewer 1

Manuscript: *ASTAF-R: An Ablation-Guided Spatio-Temporal Framework for Multi-Horizon Traffic Prediction*

In-house editor

Comment. Make the complete source code and all materials required to reproduce the experiments publicly available, and provide a direct repository link.
Response: The complete preprocessing, model, training, evaluation, compact baseline, analysis, and figure-generation code is publicly accessible at `https://github.com/shishirbit/ASTAF-R-reproducibility`. The repository includes dependency specifications, configuration values, seeds 11/22/33, data instructions, exact commands, and run metadata. The versioned reproducibility package is archived in Zenodo at `https://doi.org/10.5281/zenodo.23611484`. The manuscript Code Availability statement contains direct links to both records.

Reviewer 1

Comment 1. Ensure that stride-one 12-to-12 windows do not generate overlap leakage across split boundaries and report precise boundary management.
Response: The raw timeline is now split before any windows are constructed. Windows are generated independently inside each segment. METR-LA uses raw intervals [0,23990), [23990,27417), and [27417,34272), producing 23,967/3,404/6,832 windows. PeMS-BAY uses [0,36473), [36473,41683), and [41683,52105), producing 36,450/5,188/10,398 windows. Every run records the first and last input and target timestamps. The raw-observation intersections between training, validation, and test sets are empty.

Comment 2. Strengthen the justification for DCRNN-, STGCN-, Graph WaveNet-, AGCRN-, STID-, and Transformer-style baselines, which are compact reimplementations rather than official implementations.
Response: We retain these models strictly as compact common-code controls. The manuscript now states this limitation in the abstract, experimental setup, results, captions, conclusion, and Code Availability statement. All controls share identical windows, masks, optimizer family, stopping rule, and metrics. We do not describe them as faithful official reproductions and make no claim against the best published or officially tuned versions.

Comment 3. Clarify component selection because the two-block, top-$k = 20$ model slightly outperforms one graph block and top-$k = 40$ on one validation comparison, and learned decomposition is similar to simpler alternatives.
Response: Selection is now stated exactly. Nine principal candidates were run with three seeds on both datasets, and the frozen criterion was mean across-dataset relative validation MAE. The two-block top-$k = 40$ candidate had a relative score of 1.000691, compared with 1.000794 for two-block top-$k = 20$; the one-block candidate scored 1.003317. The difference is narrow, and the manuscript does not claim a unique optimum. Decomposition-disabled, fixed, and learned variants are all reported; their close performance is explicitly interpreted as insufficient evidence for a uniquely superior fusion rule.

Comment 4. Confirm the reported cross-seed graph correlation and top-20 Jaccard values of 1.000 with greater precision and initialization detail.

Response: The values are 1.000000000000 for all six seed pairs. A checkpoint audit revealed why: although source and target graph embeddings were initialized with seed-dependent Xavier-uniform values, both matrices have zero norm at every restored best checkpoint. All logits are therefore zero; deterministic top-k tie handling selects the same 40 entries, each with weight 0.025. We now report this as graph-parameter collapse, not remarkable adaptive-topology robustness. The title and claims no longer rely on learned-topology recovery.

Comment 5. Account more effectively for seed variability.

Response: All learned primary models now use seeds 11, 22, and 33. ASTAF-R obtains 2.995±0.008 mph on METR-LA and 1.542±0.001 mph on PeMS-BAY. We provide seed-paired differences and exact sign-flip tests, while noting that three seeds give a minimum two-sided exact p value of 0.25. We additionally aggregate errors into non-overlapping calendar-day blocks (24 days per METR-LA seed and 37 per PeMS-BAY seed) and report deterministic 10,000-repetition bootstrap intervals.

Comment 6. Discuss the trade-off between accuracy and substantially higher inference memory/latency than simpler baselines.

Response: A dedicated efficiency table now reports parameter count, profiler FLOPs, synchronized inference latency, and peak GPU memory for each dataset and model. The discussion states directly that ASTAF-R is not preferred when minimum latency or memory dominates accuracy and that the measurements are specific to the recorded laptop GPU and software environment.

Comment 7. Keep generality claims minimal because evaluation is limited to two regularly sampled California freeway-speed datasets without exogenous variables.

Response: The abstract, Results, Scope, and Conclusion now limit every claim to the controlled METR-LA and PeMS-BAY protocol. We explicitly exclude claims of universal superiority, geographic transfer, irregular-sampling robustness, exogenous-variable fusion, and calibrated uncertainty forecasting.

Summary of principal changes

- Raw timelines are split before windows; exact temporal boundaries and zero overlap are recorded.
- Ninety-four validation runs and fifty locked-test runs are reported.
- Architecture selection, locked evaluation, seed analysis, and day-block analysis are separated.
- Graph identity across seeds is explained as zero-embedding collapse.
- Accuracy–latency–memory trade-offs and baseline scope are explicit.
- The manuscript is deterministic and contains no uncertainty-performance claim.
- Complete code is public, with a Zenodo DOI to be inserted after the versioned release is archived.